The one-loop two-soliton scattering phase for the good Boussinesq equation

Project *quantum_boussinesq_solitons*, Phase 2 (next step 2)

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Abstract

We compute the one-loop ($O(\hbar^0)$) two-soliton scattering phase for the good Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ directly from the fluctuation determinant around the two-soliton background, to decide between two candidate quantum S-matrices that agree at $O(\hbar^{-1})$ and differ at $O(\hbar^0)$:

- **(A)** the Gamma candidate continued from the bad equation, whose semiclassical expansion gives $\arg S - \delta_0/\hbar \to (\pi/2)\tau \operatorname{sgn}\eta + O(\hbar)$ ($\tau = +1$ for overtaking, -1 for head-on);
- **(B)** the pure exponential $S = e^{i\delta_0/\hbar}$, with zero $O(\hbar^0)$ phase.

Our main findings are:

1. *Well-posed, reflectionless phonon scattering on the one-soliton background (§2).* Unlike the bad equation, the good equation has real phonon frequency $\omega(q) = |q|\sqrt{1+q^2}$ for all $q \in \mathbb{R}$. Linearising Hirota's τ-function gives exact reflectionless scattering eigenmodes ($R(q) = 0$ and zero asymptotic evanescent tails) with unitary transmission factor $A(1, \mathrm{ph}(q)) = (z_a/z_b)/\overline{(z_a/z_b)}$, $|A| = 1$ for all $q \neq 0$. The unwrapped phase shift $\theta(q;\alpha,\epsilon)$ decomposes into an exact three-root sum over $(\phi_1,\phi_2,\phi_3) = (\alpha + 2\pi/3, \alpha - 2\pi/3, \alpha)$ with weights $(c_1,c_2,c_3) = (+1,+1,-2)$.
2. *Levinson count and heavy-branch linear instability (§2).* The total phase winding $N_{\mathrm{Lev}} = -(\Delta\theta_+ + \Delta\theta_-)/(2\pi)$ equals $+1$ in the regular region $\alpha < \pi/3$ (counting the translational zero mode $K = k_1$) and jumps to $+2$ on the heavy branch $\alpha > \pi/3$. The extra discrete mode is an exact normalisable exponentially growing/decaying mode with co-moving frequency $\tilde{\omega} = \pm i\sqrt{3}(\sin^2\alpha - 3/4)$, arising from the secondary KP-triple roots $K_{cb} = \sin(\alpha - \pi/3)$ and $K_{ac} = \sin(\alpha + \pi/3)$ entering $(0, k_1)$ across the cusp $\alpha = \pi/3$.
3. *Closed-form dilogarithm formula for the two-soliton one-loop phase (§3).* On the two-soliton background, the linearised three-soliton τ-function proves that Faddeev–Korepin's reflection/off-diagonal operator vanishes identically ($\hat{c} = 0$). Because $\theta_1(q)\theta_2'(q) = O(q^{-3})$ as $|q| \to \infty$, the Faddeev–Korepin phonon integral is UV-convergent on $(-\infty,\infty)$ without any cutoff or counterterm. Moreover, it evaluates in closed form in terms of the Legendre chi function $\chi_2(z) = \frac{1}{2}[\mathrm{Li}_2(z) - \mathrm{Li}_2(-z)]$.
4. *Verdict on (A) vs. (B): a third answer (§4).* In the regular region ($\alpha_j < \pi/3$, $A_{12} > 0$), the one-loop phase is a *non-constant, velocity-dependent function of* (α_1,α_2). It therefore rules out **both** Candidate (A) and Candidate (B) as exact quantum S-matrices: neither contains the genuine two-body dilogarithmic phonon dressing $J_{12}(\alpha_1,\alpha_2)$. Remarkably, however:
   - in the overtaking sector ($\alpha_1 < \alpha_2 < \pi/3$), the $q = 0$ zero-mode Levinson jump between the raw Faddeev–Korepin integral I_{12}^{same} and the antisymmetrised integral J_{12}^{same} is precisely $-\pi/2$, which reproduces the $-\pi/2$ constant of Candidate (A) at $\alpha_1 \to \alpha_2$ (where $J_{12}^{\mathrm{same}} \to 0$), while adding the non-trivial two-body function $J_{12}^{\mathrm{same}}(\alpha_1,\alpha_2) \in (0,\pi/2)$;

- in the head-on sector ($\epsilon_1 = +1, \epsilon_2 = -1$), $I_{12}^{\rm head} = J_{12}^{\rm head}$ has no $\pm\pi/2$ constant and vanishes as $O(\alpha_1\alpha_2 \ln(\alpha_1\alpha_2)) \to 0$ at small amplitude, agreeing with Candidate (B) and the quantum Toda Fermi-edge limit in that corner, while becoming $O(1)$ at finite α_j.

5. *Small-amplitude limit, cusp, and resonance lines (§5–6).* As $\alpha_j \to 0$, $J_{12}^{\rm same}(\lambda\alpha_1, \lambda\alpha_2)$ converges to the scale-invariant KdV dilogarithm function $J_{\rm KdV}(\alpha_1/\alpha_2)$, with no comoving phonon capture resonance (since $|v_g(q)| > 1 > |v_j|$). At the cusp $\alpha_j \to \pi/3$, $dP_j/d\alpha_j \to 0$, the Darboux shift $\Delta X_j'$ has a simple pole, and I_{12} jumps by $\pm\pi$ as the heavy-branch instability mode detaches from $q = 0$. Across all three classical resonance lines ($A_{12} \to 0$ or ∞), the fixed-α phonon integral I_{12}, J_{12} and its first derivatives remain completely finite and smooth, whereas the classical delay ΔX_j and the momentum-renormalisation shift $\delta_{\rm renorm}$ diverge logarithmically as $\ln|A_{12}|$.

All formulas are verified to 35 digits in `phase2/scripts/oneloop_good_phonon_scattering.py`, `phase2/scripts/oneloop_good_twosoliton_phase.py`, and `phase2/scripts/oneloop_good_limits_cusp_resona` each ends with `ALL CHECKS PASSED`.

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1 Conventions and the $O(\hbar^0)$ question

We work throughout with the **good** Boussinesq equation in the fixed project conventions (`README.md`, `phase2/report.pdf`, `phase2/defect_semiclassics.pdf`):

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \phi_x = 2(\log\tau)_{xx}, \tag{1}$$

with Lagrangian density and canonical symplectic form

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \phi_x^3 - \tfrac{1}{2}\phi_{xx}^2, \qquad \pi = \phi_t, \qquad \Omega = \int_{-\infty}^{\infty} dx\, d\pi \wedge d\phi. \tag{2}$$

Planck's constant $\hbar$ refers to this canonical normalisation.

One- and two-soliton kinematics. A single soliton with angle parameter $\alpha \in (0, \pi/2)$ and direction $\epsilon \in \{+1, -1\}$ has **(a)**/**(c)**:

$$u_0(x,t) = \frac{k^2}{2}\operatorname{sech}^2\!\left(\frac{k(x-vt-x_0)}{2}\right), \qquad k = \sin\alpha, \qquad v = \epsilon\cos\alpha, \qquad \ell = -\epsilon\cos 3\alpha,$$
$$M = \int_{-\infty}^{\infty} u_0\,dx = 2k, \qquad \Pi = \int_{-\infty}^{\infty} \pi_0\,dx = -Mv = -\epsilon\sin 2\alpha, \tag{3}$$
$$P = \tfrac{2}{3}\epsilon k^3\cos\alpha, \qquad \frac{dP}{d\alpha} = \tfrac{2}{3}\epsilon\sin\alpha\sin 3\alpha, \qquad E = \tfrac{1}{3}k^3(2\cos^2\alpha + 1), \qquad \frac{dE}{d\alpha} = v\frac{dP}{d\alpha}.$$

Its two KP roots are

$$p_a = \frac{1}{2}\left(k + \frac{\epsilon\cos\alpha}{\sqrt{3}}\right), \qquad p_b = \frac{1}{2}\left(\frac{\epsilon\cos\alpha}{\sqrt{3}} - k\right), \tag{4}$$

which belong to the three-root KP triple $\{p_a, p_b, p_c\}$ of $\ell(p) := -3\sqrt{3}\,p(1-4p^2) = \ell$, where the third root is $p_c = -p_a - p_b = -\epsilon\cos\alpha/\sqrt{3}$ **(c)** (`phase2/defect_semiclassics.pdf`).

For two solitons with particle 1 starting on the left at $t \to -\infty$ ($v_1 > v_2$) and ending on the right at $t \to +\infty$, the Hirota phase factor and semiclassical two-soliton phase are **(a)**/**(c)**:

$$A_{12} = \frac{(p_{1a}-p_{2a})(p_{1b}-p_{2b})}{(p_{1a}-p_{2b})(p_{1b}-p_{2a})}, \qquad \delta_0 = \tfrac{2}{9}(\ell_1-\ell_2)\ln A_{12} + T_{12}, \qquad T_{12} = \tfrac{2}{3}M_1M_2(v_1-v_2) = -\tfrac{2}{3}(\Pi_1M_2 - \Pi_2M_1). \tag{5}$$

The topological cross term T_{12} follows from exact symplecticity of the asymptotic two-soliton phase space with ordering-dependent cross terms $d\Pi_i \wedge dM_j$; in the Darboux coordinates $X_j' = X_j + \gamma_j \sum_{i \text{ left of } j} M_i$, where

$$\gamma_j = \frac{d\Pi_j/d\alpha_j}{dP_j/d\alpha_j} = -\frac{3\cos 2\alpha_j}{\sin\alpha_j \sin 3\alpha_j}, \tag{6}$$

the canonical spatial shifts are $\Delta X_1' = -\ln(A_{12})/k_1 + \gamma_1 M_2 = -\partial\delta_0/\partial P_1$ and $\Delta X_2' = +\ln(A_{12})/k_2 - \gamma_2 M_1 = -\partial\delta_0/\partial P_2$ **(c)** (`phase2/report.pdf`, `phase2/defect_semiclassics.pdf`). The regular region is $\alpha_1, \alpha_2 < \pi/3$ with $A_{12} > 0$. At the cusp $\alpha = \pi/3$, $dP/d\alpha = 0$. Every breather of the good equation is singular **(a)**/**(c)**.

The two candidate S-matrices. Let $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and $R(x) = \Gamma(1+ix)/\Gamma(1-ix)$.

- **(A) The Gamma candidate**, continued from the bad equation (`phase2/smatrix_ansatz.pdf`, `phase2/defect_semiclassics.pdf`):

$$S_{\mathrm{A}}^{\mathrm{same}} = R(\eta)\left(\frac{e^2 A_{12}}{\eta^2}\right)^{i\eta} e^{iT_{12}/\hbar}, \qquad S_{\mathrm{A}}^{\mathrm{head}} = R(\eta)^{-1}\left(\frac{A_{12}\eta^2}{e^2}\right)^{i\eta} e^{iT_{12}/\hbar}. \tag{7}$$

  By Stirling's expansion $R(\eta)(e/\eta)^{2i\eta} = \exp\left(i\frac{\pi}{2}\operatorname{sgn}\eta - \frac{i}{6\eta} + O(\eta^{-3})\right)$ as $|\eta| \to \infty$, Candidate (A) predicts at one loop ($O(\hbar^0)$):

$$\delta_1^{(\mathrm{A})} := \lim_{\hbar\to 0}\left(\arg S_{\mathrm{A}} - \frac{\delta_0}{\hbar}\right) = \frac{\pi}{2}\tau\operatorname{sgn}\eta, \qquad \tau = \begin{cases} +1, & \text{same direction } (\epsilon_1 = \epsilon_2), \\ -1, & \text{head-on } (\epsilon_1 = +1, \epsilon_2 = -1). \end{cases} \tag{8}$$

  In the regular region $\alpha_j < \pi/3$, a colliding overtaking pair ($\epsilon_1 = \epsilon_2 = +1$, $v_1 > v_2 \implies \alpha_1 < \alpha_2$) has $\ell_1 - \ell_2 = \cos 3\alpha_2 - \cos 3\alpha_1 < 0$, so $\operatorname{sgn}\eta = -1$ and $\delta_1^{(\mathrm{A})} = -\pi/2$. A regular head-on pair ($\epsilon_1 = +1, \epsilon_2 = -1$, $\alpha_1 + \alpha_2 < \pi/3$) has $\ell_1 - \ell_2 = -\cos 3\alpha_1 - \cos 3\alpha_2 = -2\cos\frac{3(\alpha_1+\alpha_2)}{2}\cos\frac{3(\alpha_1-\alpha_2)}{2} < 0$, so again $\operatorname{sgn}\eta = -1$ and $\delta_1^{(\mathrm{A})} = +\pi/2$.

- **(B) The pure exponential** $S_{\mathrm{B}} = e^{i\delta_0/\hbar}$, which predicts

$$\delta_1^{(\mathrm{B})} = 0. \tag{9}$$

2 Task 1: Linearisation about the one-soliton and phonon scattering

2.1 The fourth-order fluctuation operator and evanescent modes

Expanding $\phi = \phi_0 + \sqrt{\hbar}\,\delta\phi$ ($\delta u = \partial_x \delta\phi$) around a classical background $u_0(x,t) = \partial_x \phi_0(x,t)$ yields the quadratic fluctuation action $S^{(2)} = -\frac{1}{2}\int dt\,dx\,\delta\phi\,H\delta\phi$, where

$$H = \partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x u_0(x,t)\partial_x, \qquad H_u\,\delta u := \left[\partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x^2 u_0(x,t)\right]\delta u = 0. \tag{10}$$

Unlike KdV (first-order in ∂_t), H is *second-order in time* with a canonical kinetic term $\frac{1}{2}(\delta\phi_t)^2$ and a positive-definite fourth-order spatial principal symbol $+\partial_x^4$. On the vacuum $u_0 = 0$, plane waves $e^{iqx-i\omega t}$ satisfy $\omega^2 = q^2 + q^4$, so $\omega(q) = |q|\sqrt{1+q^2} > 0$ is real for all $q \in \mathbb{R}\setminus\{0\}$.

In the co-moving frame $\xi = x - v_1 t$ of a one-soliton $u_0(\xi)$, stationary modes $\delta u(\xi,t) = e^{-i\tilde\omega t} f(\xi)$ at co-moving frequency $\tilde\omega = \omega - v_1 q$ satisfy the fourth-order ordinary differential equation

$$\left[\partial_\xi^4 - (1-v_1^2)\partial_\xi^2 - 2iv_1\tilde\omega\partial_\xi - \tilde\omega^2 + 6\partial_\xi^2 u_0(\xi)\right] f(\xi) = 0. \tag{11}$$

As $\xi \to \pm\infty$, $u_0(\xi) \to 0$ exponentially and characteristic exponents $f \sim e^{iq\xi}$ obey the quartic dispersion relation

$$P_{\rm vac}(q;\tilde\omega) := q^4 + k_1^2 q^2 - 2v_1\tilde\omega q - \tilde\omega^2 = 0, \qquad k_1^2 = 1 - v_1^2 = \sin^2\alpha_1 > 0. \tag{12}$$

Proposition 2.1 (**(c)**, check 1 of `phase2/scripts/oneloop_good_phonon_scattering.py`)**.** *For every $\alpha_1 \in (0,\pi/2)$ and every real co-moving frequency $\tilde\omega \neq 0$, the quartic (12) has* strictly two real roots $q_+(\tilde\omega) > 0 > q_-(\tilde\omega)$ *(propagating waves) and* one complex-conjugate pair of non-real roots $q_{\rm ev}(\tilde\omega), \overline{q_{\rm ev}(\tilde\omega)}$ *with* $\operatorname{Im} q_{\rm ev}(\tilde\omega) > 0$ *(evanescent waves decaying as $\xi \to +\infty$ and $\xi \to -\infty$, respectively).*

Proof. For real q, $P''_{\rm vac}(q) = 12q^2 + 2k_1^2 > 0$ everywhere, so $P_{\rm vac}(q;\tilde\omega)$ is strictly convex on $\mathbb{R}$. Since $P_{\rm vac}(0;\tilde\omega) = -\tilde\omega^2 < 0$ and $P_{\rm vac}(\pm\infty;\tilde\omega) = +\infty$, it crosses zero exactly once on $(0,\infty)$ and once on $(-\infty,0)$. The remaining two roots of the real quartic must be a complex-conjugate non-real pair. □

2.2 Exact reflectionlessness and unitarity via Hirota τ-linearisation

A generic fourth-order scattering problem at frequency $\tilde\omega$ has a 2×2 unitary scattering matrix connecting the two propagating channels $q_\pm(\tilde\omega)$, constructed by matching four boundary conditions (including the decaying evanescent mode at each end). For the integrable Boussinesq one-soliton, linearising the two-soliton Hirota τ-function solves this fourth-order boundary-value problem in closed form.

In Hirota's two-soliton solution $\tau = 1 + e^{\theta_1} + \varepsilon(e^{\theta_2} + A_{12}e^{\theta_1+\theta_2})$ with $\theta_j = K_j x - \Omega_j t$ ($\Omega_j^2 = K_j^2 - K_j^4$), setting $K_2 = iq$ ($q \in \mathbb{R}\setminus\{0\}$) and $\Omega_2 = i\omega(q)$ with $\omega(q) = |q|\sqrt{1+q^2} > 0$ corresponds to continuing $\alpha_2 \to \alpha_{\rm ph}(q) := i\,\operatorname{arcsinh} q$ with $\epsilon_{\rm ph} = \operatorname{sgn}(q)$, so that $\sin\alpha_{\rm ph} = iq$ and $\epsilon_{\rm ph}\cos\alpha_{\rm ph} = \operatorname{sgn}(q)\sqrt{1+q^2} = \omega(q)/q$ **(c)**. Expanding $u = 2(\log\tau)_{xx}$ to $O(\varepsilon)$ gives

$$\delta u(x,t;q) = 2\partial_x^2\left[\frac{1 + A(1,\mathrm{ph}(q))e^{\theta_1}}{1+e^{\theta_1}}\,e^{iqx-i\omega(q)t}\right], \qquad \delta\phi(x,t;q) = 2\partial_x\left[\frac{1 + A(1,\mathrm{ph}(q))e^{\theta_1}}{1+e^{\theta_1}}\,e^{iqx-i\omega(q)t}\right]. \tag{13}$$

Proposition 2.2 (**(c)**, checks 2–3 of `phase2/scripts/oneloop_good_phonon_scattering.py`)**.** *For all $\alpha_1 \in (0,\pi/2)$, $\epsilon_1 \in \{+1,-1\}$, and $q \in \mathbb{R}\setminus\{0\}$:*

1. *The mode (13) is an exact solution of the fourth-order linearised equation $H_u\,\delta u = 0$ (verified to $< 10^{-35}$).*

2. *As* $x \to -\infty$ *(*$e^{\theta_1} \to 0$*),* $\delta u(x,t;q) \to -2q^2 e^{iqx - i\omega(q)t}$*; as* $x \to +\infty$ *(*$e^{\theta_1} \to \infty$*),* $\delta u(x,t;q) \to -2q^2 A(1,\mathrm{ph}(q)) e^{iqx - i\omega(q)t}$*. Thus* phonon scattering on the one-soliton is strictly reflectionless ($R(q) = 0$) and has zero asymptotic evanescent tails ($C_\pm(q) = 0$).

3. *Under the continuation* $\alpha_{\mathrm{ph}} = i \operatorname{arcsinh} q$, $\epsilon_{\mathrm{ph}} = \mathrm{sgn}(q)$*, the phonon's KP roots* (4) *are complex conjugates:*

$$p_{\mathrm{ph,a}}(q) = \frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} + \frac{iq}{2}, \qquad p_{\mathrm{ph,b}}(q) = \overline{p_{\mathrm{ph,a}}(q)} = \frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} - \frac{iq}{2}. \tag{14}$$

Since the soliton's KP roots p_{1a}, p_{1b} *are real, the transmission coefficient* $T(q) = A(1,\mathrm{ph}(q))$ *takes the manifestly unitary form*

$$A(1,\mathrm{ph}(q)) = \frac{(p_{1a} - p_{\mathrm{ph,a}})(p_{1b} - \overline{p_{\mathrm{ph,a}}})}{(p_{1a} - \overline{p_{\mathrm{ph,a}}})(p_{1b} - p_{\mathrm{ph,a}})} = \frac{z_a(q)/z_b(q)}{\overline{z_a(q)/z_b(q)}}, \qquad z_{a,b}(q) := \left(p_{1a,b} - \frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}}\right) - \frac{iq}{2}. \tag{15}$$

Consequently, $|A(1,\mathrm{ph}(q))| = 1$ *for* all $q \in \mathbb{R} \setminus \{0\}$*: unlike the bad Boussinesq equation (*`phase2/oneloop.pdf`*), the good equation has no unstable continuum phonon band.*

2.3 Closed-form transmission phase shift $\theta(q;\alpha,\epsilon)$

Because $z_a(q) - z_b(q) = p_{1a} - p_{1b} = k_1 > 0$ is real and positive, we have

$$\frac{z_a(q)}{z_b(q)} = 1 + \frac{k_1}{z_b(q)}, \qquad \mathrm{Im}\left(\frac{z_a(q)}{z_b(q)}\right) = \frac{k_1 q}{2|z_b(q)|^2}. \tag{16}$$

For $q \neq 0$, $\mathrm{Im}(z_a(q)/z_b(q))$ has the strict sign of q and never vanishes! Thus $z_a(q)/z_b(q)$ lies strictly in the upper half-plane for all $q > 0$ and strictly in the lower half-plane for all $q < 0$, and approaches 1 as $q \to \pm\infty$. This defines a unique, smooth, unwrapped transmission phase shift $\theta(q;\alpha,\epsilon)$ on $(-\infty,0)$ and $(0,\infty)$ vanishing at $q = \pm\infty$:

$$A(1,\mathrm{ph}(q)) = e^{i\theta(q;\alpha,\epsilon)}, \qquad \theta(q;\alpha,\epsilon) := 2\,\mathrm{atan2}\left(\mathrm{Im}\left(1 + \frac{k_1}{z_b(q)}\right), \mathrm{Re}\left(1 + \frac{k_1}{z_b(q)}\right)\right) \in (-2\pi, 2\pi). \tag{17}$$

Proposition 2.3 (**(c)**, check 3 of `phase2/scripts/oneloop_good_phonon_scattering.py`)**.** *Let* $(\phi_1,\phi_2,\phi_3) := (\alpha + \frac{2\pi}{3}, \alpha - \frac{2\pi}{3}, \alpha)$ *and* $(c_1,c_2,c_3) := (+1,+1,-2)$*. For all* $\alpha \in (0,\pi/2)$*,* $\epsilon \in \{+1,-1\}$*, and* $q \in \mathbb{R} \setminus \{0\}$*, the unwrapped phase shift* (17) *decomposes into the exact three-root sum*

$$\boxed{\theta(q;\alpha,\epsilon) = -\sum_{r=1}^{3} c_r \left[\mathrm{sgn}(q)\arctan\left(\frac{\sin\phi_r}{|q|}\right) + \epsilon \arctan\left(\frac{\sin\phi_r\cos\phi_r}{|q|\sqrt{1+q^2} + q^2 + \sin^2\phi_r}\right)\right]} \tag{18}$$

where every arctan *takes its principal value in* $(-\pi/2,\pi/2)$*. Its derivative* $\theta'(q;\alpha,\epsilon) = \partial_q\theta(q;\alpha,\epsilon)$ *decomposes into an* ϵ*-independent even rational part and an* ϵ*-odd algebraic part:*

$$\boxed{\frac{\partial\theta(q;\alpha,\epsilon)}{\partial q} = \sum_{r=1}^{3} \frac{c_r \sin\phi_r}{q^2 + \sin^2\phi_r} + \frac{\epsilon\,\mathrm{sgn}(q)}{\sqrt{1+q^2}} \sum_{r=1}^{3} \frac{c_r \sin\phi_r\cos\phi_r}{q^2 + \sin^2\phi_r}.} \tag{19}$$

In the hyperbolic variable $q = \sinh u$ *(*$u > 0$*,* $\epsilon = +1$*),* $\partial_u\theta(\sinh u;\alpha,+1) = \sum_{r=1}^{3} \frac{c_r \sin\phi_r}{\cosh u - \cos\phi_r}$ *and* $\partial_u\theta(-\sinh u;\alpha,+1) = \sum_{r=1}^{3} \frac{c_r \sin\phi_r}{\cosh u + \cos\phi_r}$*. As* $|q| \to \infty$*, the trigonometric identities* $\sum_{r=1}^{3} c_r \sin\phi_r \cos^m\phi_r = (-3\sin\alpha, -\frac{3}{2}\sin 2\alpha, -\frac{3}{4}\sin\alpha)$ *for* $m = 0,1,2$ *yield the Laurent expansion*

$$\theta(q;\alpha,\epsilon) = \frac{3k_1}{q} + \frac{3k_1 v_1}{2q^2} - \frac{3k_1}{4q^3} + O(q^{-4}), \qquad \frac{\partial\theta(q;\alpha,\epsilon)}{\partial q} = -\frac{3k_1}{q^2} - \frac{3k_1 v_1}{q^3} + \frac{9k_1}{4q^4} + O(q^{-5}). \tag{20}$$

2.4 Levinson-type mode count, zero modes, and heavy-branch instability

Because the odd part of (19) cancels between $+q$ and $-q$, the total phase winding over the continuum $(-\infty, 0) \cup (0, \infty)$ is the integral of three elementary Lorentzians:

$$\Delta\theta_{\rm tot}(\alpha) := \int_{-\infty}^{0} \theta'(q)\, dq + \int_{0}^{\infty} \theta'(q)\, dq = \int_{0}^{\infty} \sum_{r=1}^{3} \frac{2c_r \sin\phi_r}{q^2 + \sin^2\phi_r}\, dq = \pi \sum_{r=1}^{3} c_r \,\mathrm{sgn}(\sin\phi_r). \quad (21)$$

For $\alpha \in (0, \pi/2)$, we have $\sin\phi_3 = \sin\alpha > 0$ and $\sin\phi_2 = \sin(\alpha - 2\pi/3) < 0$, whereas $\sin\phi_1 = \sin(\alpha + 2\pi/3) = \sin(\pi/3 - \alpha)$ is *positive for $\alpha < \pi/3$ and negative for $\alpha > \pi/3$*!

Proposition 2.4 (**(c)**, checks 4–5 of `phase2/scripts/oneloop_good_phonon_scattering.py`)**.** *Define the branch windings $\Delta\theta_+ = \theta(+\infty) - \theta(0^+) = -\theta(0^+)$ and $\Delta\theta_- = \theta(0^-) - \theta(-\infty) = \theta(0^-)$, and the Levinson mode count $N_{\rm Lev} := -(\Delta\theta_+ + \Delta\theta_-)/(2\pi) = [\theta(0^+) - \theta(0^-)]/(2\pi)$.*

1. Regular branch $\alpha \in (0, \pi/3)$: *For $\epsilon = +1$, $\theta(0^+) = 2\pi$ and $\theta(0^-) = 0$ (for $\epsilon = -1$, $\theta(0^+) = 0$ and $\theta(0^-) = -2\pi$). Thus $\Delta\theta_{\rm tot} = -2\pi$ and*

$$N_{\rm Lev} = 1 \qquad (\alpha < \pi/3). \quad (22)$$

2. Heavy branch $\alpha \in (\pi/3, \pi/2)$: *For both $\epsilon = \pm 1$, $\theta(0^+) = 2\pi$ and $\theta(0^-) = -2\pi$, so $\Delta\theta_+ = \Delta\theta_- = -2\pi$, $\Delta\theta_{\rm tot} = -4\pi$, and*

$$N_{\rm Lev} = 2 \qquad (\alpha > \pi/3). \quad (23)$$

3. Discrete spectrum from KP triples: *A continuation $q = -iK$ ($\mathrm{Re}\, K > 0$) of (13) yields a spatially normalisable discrete mode on $\mathbb{R}$ iff $A(1, K) = 0$ with $0 < \mathrm{Re}\, K < k_1$, or $K = k_1$ where ∂_ξ^2 annihilates the constant 1 in $1 + A(1, K)e^{k_1\xi}$. By (15), $A(1, K) = 0$ iff the perturbation's KP root pair (p_{2a}, p_{2b}) is drawn from the soliton's own KP triple $\{p_{1a}, p_{1b}, p_{1c}\}$:*

   - *(p_{1a}, p_{1b}) gives $K = k_1$ and $\tilde\omega = 0$: the* translational zero mode *$\delta u_0(\xi) = \partial_\xi u_0(\xi)$ (present for all $\alpha \in (0, \pi/2)$, together with its secular partner $\partial_P u_0$, which degenerates at the cusp $dP/d\alpha = 0$).*
   - *(p_{1c}, p_{1b}) and (p_{1a}, p_{1c}) give wavenumbers $K_{cb} = p_{1c} - p_{1b} = \sin(\alpha - \pi/3)$ and $K_{ac} = p_{1a} - p_{1c} = \sin(\alpha + \pi/3) = k_1 - K_{cb}$, with purely imaginary co-moving frequencies $\tilde\omega = \pm i\gamma(\alpha)$, where*

$$\gamma(\alpha) = \sqrt{3}\left(\sin^2\alpha - \tfrac{3}{4}\right), \quad (24)$$

   and exact spatial profile $\delta u_\pm(\xi, t) = e^{\pm\gamma(\alpha)t}\partial_\xi^2\left[\frac{e^{K_{cb}\xi}}{1+e^{k_1\xi}}\right]$. For $\alpha < \pi/3$, $K_{cb} < 0$ and $K_{ac} > k_1$, so $\delta u_\pm$ blows up exponentially as $\xi \to -\infty$ and there are no bound or unstable modes beyond the translational zero mode *($N_{\rm Lev} = 1$). For $\alpha > \pi/3$, $0 < K_{cb}, K_{ac} < k_1$ and $\gamma(\alpha) > 0$, so $\delta u_\pm$ are* normalisable exponentially growing and decaying linear instability modes *($N_{\rm Lev} = 2$), representing the linearised resonant Y-junction decay of a heavy soliton into the two light solitons of its KP triple.*

3 Task 2: Two-soliton $O(\hbar^0)$ phase on the two-soliton background

3.1 Exact two-soliton phonon eigenmodes and $\hat c = 0$

Following Faddeev and Korepin (FK) [1] (§5.4 and Appendices 5–6, full text read), the one-loop correction to the two-soliton S-matrix is governed by the scattering of continuum modes on

the time-dependent two-soliton background $u_{12}(x,t) = 2(\log\tau_{12})_{xx}$, where $\tau_{12} = 1 + E_1 + E_2 + A_{12}E_1E_2$ with $E_j = \exp(k_j(x - v_jt - x_{0j}))$.

In a general non-integrable theory, a plane wave $e^{iqx-i\omega(q)t}$ scattering through the two-soliton collision region emerges at $t \to -\infty$ as a superposition involving an off-diagonal integral kernel $\hat{c}(q,q')$, which adds $-\frac{1}{2}\operatorname{Tr}\ln(1+\hat{c})$ to $\ln S_0$ **(a)** [1]. For the good Boussinesq equation, linearising the three-soliton Hirota τ-function in the third (phonon) amplitude $K_3 = iq$, $\Omega_3 = i\omega(q)$ gives the exact time-dependent eigenmode

$$\delta u_{12}(x,t;q) = 2\partial_x^2\left[\frac{1 + A(1,\mathrm{ph}(q))E_1 + A(2,\mathrm{ph}(q))E_2 + A_{12}A(1,\mathrm{ph}(q))A(2,\mathrm{ph}(q))E_1E_2}{1 + E_1 + E_2 + A_{12}E_1E_2}\, e^{iqx-i\omega(q)t}\right]. \tag{25}$$

Proposition 3.1 (**(c)**, check 1 of `phase2/scripts/oneloop_good_twosoliton_phase.py`)**.** *For every regular two-soliton background ($\alpha_1, \alpha_2 \in (0, \pi/3)$, $A_{12} > 0$) and every $q \in \mathbb{R} \setminus \{0\}$, the function (25) satisfies the time-dependent fourth-order linearised equation $\left[\partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x^2 u_{12}(x,t)\right]\delta u_{12} = 0$ identically for all $(x,t) \in \mathbb{R}^2$ (verified at finite collision times to $< 10^{-36}$). It is purely monochromatic of wavenumber q as $t \to \pm\infty$ and $x \to \pm\infty$, with zero reflection and exact factorised transmission amplitude $A_{12}^{\mathrm{ph}}(q) = A(1,\mathrm{ph}(q))A(2,\mathrm{ph}(q)) = e^{i\theta_1(q)+i\theta_2(q)}$. Therefore, the Faddeev–Korepin off-diagonal operator vanishes identically: $\hat{c} = 0$.*

3.2 The Faddeev–Korepin phonon sum, UV convergence, and closed-form dilogarithm solution

Because $H = \partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x u_0(x,t)\partial_x$ is second-order in time with unit coefficient of ∂_t^2, the Wronskian and Krein spectral-density identities of FK [1] (§5.4, eqs. 4.14–4.28 and Appendix 6, eq. A.6.2) apply verbatim. Specifically, with particle 1 starting on the left at $t' \to -\infty$ and ending on the right at $t'' \to +\infty$:

- Differentiating the negative-frequency Jost mode $\psi_-(x,t;q)$ (normalised to $e^{iqx-i\omega(q)t}$ as $x \to -\infty$) with respect to α_1 gives a non-zero result only near soliton 1 and to its right. At $t'' \to +\infty$, soliton 1 is to the right of soliton 2, so the region to the right of soliton 1 is vacuum and cancels against the single-soliton determinant $\det H(u_1)$ in the denominator. At $t' \to -\infty$, soliton 2 lies to the right of soliton 1, where $\partial_{\alpha_1}\ln\psi_- = i\,\partial_{\alpha_1}\theta_1(q;\alpha_1,\epsilon_1)$, and integrating across soliton 2 against the Krein density $-2i\,\partial_q\theta_2(q;\alpha_2,\epsilon_2)$ divided by the Wronskian $-4\pi i$ (times $-\frac{1}{2}$ from $\det^{-1/2}$) gives FK eq. (4.27):

$$\frac{\partial\delta_{\mathrm{ph}}(\alpha_1,\epsilon_1;\alpha_2,\epsilon_2)}{\partial\alpha_1} = \frac{1}{4\pi}\int_{-\infty}^{\infty} dq\,\frac{\partial\theta_1(q;\alpha_1,\epsilon_1)}{\partial\alpha_1}\frac{\partial\theta_2(q;\alpha_2,\epsilon_2)}{\partial q}. \tag{26}$$

- Differentiating with respect to α_2 receives its non-cancelling contribution from the upper limit $t'' \to +\infty$ (where soliton 2 is on the left and soliton 1 lies to its right), giving the opposite sign:

$$\frac{\partial\delta_{\mathrm{ph}}(\alpha_1,\epsilon_1;\alpha_2,\epsilon_2)}{\partial\alpha_2} = -\frac{1}{4\pi}\int_{-\infty}^{\infty} dq\,\frac{\partial\theta_2(q;\alpha_2,\epsilon_2)}{\partial\alpha_2}\frac{\partial\theta_1(q;\alpha_1,\epsilon_1)}{\partial q}. \tag{27}$$

UV convergence and the $q = 0$ Levinson boundary term (I_{12} vs. J_{12}). By (20), $\theta_1(q)\theta_2'(q) = -9k_1k_2/q^3 + O(q^{-4})$ as $|q| \to \infty$, so the integrals (26)–(27) as well as the undifferentiated integrals

$$I_{12} := \frac{1}{4\pi}\int_{-\infty}^{\infty} dq\,\theta_1(q)\theta_2'(q), \qquad J_{12} := \frac{1}{8\pi}\int_{-\infty}^{\infty} dq\,\left[\theta_1(q)\theta_2'(q) - \theta_2(q)\theta_1'(q)\right] \tag{28}$$

are *absolutely UV-convergent on $(-\infty,\infty)$ without any cutoff or counterterm* **(c)**. (Any local tadpole counterterm $\propto \int u\,dx$ is proportional to the conserved topological charge $M = \int u\,dx$ and cancels identically in the ratio $\det H(u_{12})/[\det H(u_1)\det H(u_2)]$ since $M_{12} = M_1 + M_2$.)

On $(-\infty,0)\cup(0,\infty)$, integrating I_{12} by parts yields a boundary term at $q=0^{\pm}$ from the 2π Levinson jump of $\theta_j(q)$ (Proposition 2.4):

$$I_{12}+I_{21}=\frac{1}{4\pi}\Big[\theta_1(0^-)\theta_2(0^-)-\theta_1(0^+)\theta_2(0^+)\Big]=\begin{cases}-\pi, & \epsilon_1=\epsilon_2 \text{ (same direction)},\\ 0, & \epsilon_1=-\epsilon_2 \text{ (head-on)},\end{cases}\qquad(\alpha_1,\alpha_2<\pi/3). \tag{29}$$

Because the values $\theta_j(0^{\pm})\in\{0,\pm 2\pi\}$ are topological constants independent of $\alpha_j\in(0,\pi/3)$, we have $\partial_{\alpha_j}\theta_j(0^{\pm})=0$. Hence integrating (26)–(27) by parts in q produces *zero* boundary term at $q=0$, proving that I_{12} and J_{12} share the *exact same gradient* on $(0,\pi/3)^2$:

$$\nabla_\alpha I_{12}=\nabla_\alpha J_{12}=\left(\frac{\partial\delta_{\rm ph}}{\partial\alpha_1},\frac{\partial\delta_{\rm ph}}{\partial\alpha_2}\right),\qquad I_{12}=J_{12}-\frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}\qquad(\alpha_1,\alpha_2<\pi/3). \tag{30}$$

In FK eq. (4.28), this $O(\hbar^0)$ integration constant is the overall factor $Z=e^{i\delta_{\rm const}}$. Following the PI's decision, we report both I_{12} and J_{12} in parallel.

Proposition 3.2 (**(c)**, check 2 of `phase2/scripts/oneloop_good_twosoliton_phase.py`)**.** *Define the odd function $L:\mathbb{R}\to\mathbb{R}$ via the Legendre chi function $\chi_2(z)=\sum_{n=0}^{\infty}\frac{z^{2n+1}}{(2n+1)^2}=\frac{1}{2}[\mathrm{Li}_2(z)-\mathrm{Li}_2(-z)]$:*

$$L(z):=\int_0^z\frac{2\ln|t|}{1-t^2}\,dt=-\operatorname{sgn}(z)\left\{2\operatorname{Re}\chi_2(|z|)+\ln|z|\,\ln\left|\frac{1-|z|}{1+|z|}\right|\right\},\qquad L(0)=0,\quad L(\pm1)=\mp\frac{\pi^2}{4}, \tag{31}$$

which satisfies the inversion identity $L(z)+L(1/z)=-\operatorname{sgn}(z)\frac{\pi^2}{2}$ for all $z\neq 0$. Let $\phi_{j,r}\in(\alpha_j+\frac{2\pi}{3},\alpha_j-\frac{2\pi}{3},\alpha_j)$ with weights $(c_1,c_2,c_3)=(+1,+1,-2)$. Then for all $\alpha_1,\alpha_2\in(0,\pi/3)$:

1. *For **same-direction (overtaking)** collisions ($\epsilon_1=\epsilon_2=+1$):*

$$\boxed{I_{12}^{\rm same}(\alpha_1,\alpha_2)=\frac{1}{2\pi}\sum_{r=1}^{3}\sum_{s=1}^{3}c_rc_s\,L\left(\frac{\tan(\phi_{1,r}/2)}{\tan(\phi_{2,s}/2)}\right),\qquad J_{12}^{\rm same}(\alpha_1,\alpha_2)=\frac{\pi}{2}+I_{12}^{\rm same}(\alpha_1,\alpha_2).} \tag{32}$$

2. *For **head-on** collisions ($\epsilon_1=+1,\epsilon_2=-1$):*

$$\boxed{I_{12}^{\rm head}(\alpha_1,\alpha_2)=J_{12}^{\rm head}(\alpha_1,\alpha_2)=\frac{1}{2\pi}\sum_{r=1}^{3}\sum_{s=1}^{3}c_rc_s\,L\left(\tan\frac{\phi_{1,r}}{2}\,\tan\frac{\phi_{2,s}}{2}\right)=I_{12}^{\rm same}(\alpha_1,\pi-\alpha_2).} \tag{33}$$

3. *The partial derivatives $\partial_{\alpha_j}I_{12}=\partial_{\alpha_j}J_{12}$ are purely elementary functions of $\sin\phi_{j,r},\cos\phi_{j,r}$, and $\ln|\tan(\phi_{j,r}/2)|$, given by $\partial_{\alpha_1}I_{12}=\epsilon_2\,\partial_{\alpha_1}K(\alpha_1,\alpha_2)-\epsilon_1\,\partial_{\alpha_1}K(\alpha_2,\alpha_1)$ with*

$$\frac{\partial K(\alpha_1,\alpha_2)}{\partial\alpha_1}=\frac{1}{2\pi}\sum_{r=1}^{3}\sum_{s=1}^{3}\frac{c_rc_s\sin\phi_{2,s}}{\sin^2\phi_{2,s}-\sin^2\phi_{1,r}}\left(\cos\phi_{2,s}\ln\left|\tan\frac{\phi_{1,r}}{2}\right|-\cos\phi_{1,r}\ln\left|\tan\frac{\phi_{2,s}}{2}\right|\right). \tag{34}$$

Proof. Splitting $\theta_j(q)=\theta_j^{\rm odd}(q)+\epsilon_j\theta_j^{\rm even}(q)$ via (18)–(19), only the even-integrand cross terms survive integration over $(-\infty,\infty)$, reducing I_{12} to $K(\alpha_1,\alpha_2):=\frac{1}{2\pi}\int_0^\infty\theta_1^{\rm odd}(q)(\theta_2^{\rm even})'(q)\,dq$ and $K(\alpha_2,\alpha_1)$. Differentiating the elementary integral $g(\kappa_1,\kappa_2)=\int_0^\infty\frac{dq}{\sqrt{1+q^2}}\frac{\arctan(\kappa_1/q)}{q^2+\kappa_2^2}$ with respect to $\kappa_1=\sin\phi_{1,r}$ replaces $\arctan(\kappa_1/q)$ by $q/(q^2+\kappa_1^2)$, and the substitution $t=\sqrt{1+q^2}\in[1,\infty)$ ($dt=q\,dq/\sqrt{1+q^2}$) yields (34). Passing to $x=\tan(\phi_{1,r}/2)$ and $y=\tan(\phi_{2,s}/2)$ decomposes the differential into $\frac{1}{4}[dL(x/y)-dL(xy)]$, and applying $L(u)+L(1/u)=-\operatorname{sgn}(u)\frac{\pi^2}{2}$ together with $\sum_{r=1}^{3}c_r\operatorname{sgn}(\sin\phi_{j,r})=-2$ on $(0,\pi/3)$ gives (32)–(33). Every step is verified against 35-digit numerical quadrature to $<5\times10^{-36}$. □

3.3 Maslov index, zero modes, topological cross terms, and parameter renormalisation

Proposition 3.3 (**(c)**, checks 3–4 of `phase2/scripts/oneloop_good_twosoliton_phase.py`)**.** *In the regular region $\alpha_1, \alpha_2 \in (0, \pi/3)$ with $A_{12} > 0$:*

1. Zero modes and wave-packet Maslov index: *Each one-soliton has a single translational zero mode and no oscillating bound states ($N_{\mathrm{Lev}} = 1$). In the collective-coordinate reduction formula (FK [1], Appendix 1, eq. A.1.9 and Appendix 5, eq. A.5.26), the zero-mode Wronskian is time-independent and cancels against the single-soliton propagators in the denominator. The stationary-phase Hessian of the momentum wave-packet integral has eigenvalues $(t''-t')\frac{dv_j}{dP_j} - \frac{\partial \Delta X'_j}{\partial P_j} = (t''-t')\frac{dv_j}{dP_j}[1+O((t''-t')^{-1})]$ as $t''-t' \to +\infty$, which has the* exact same signature *as the free single-soliton propagators $(t''-t')\frac{dv_j}{dP_j}$ for all $\alpha_j \neq \pi/3$ (where $\frac{dv_j}{dP_j} = -\frac{3}{2\sin 3\alpha_j} < 0$ is finite and non-zero). Hence the relative wave-packet Maslov index vanishes in the regular region:*

$$\mu_{\mathrm{Maslov}} = 0 \qquad (\alpha_1, \alpha_2 < \pi/3). \tag{35}$$

2. Topological cross terms at $O(\hbar^0)$: *The classical cross term $T_{12} = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1) = \frac{2}{3} M_1 M_2 (v_1 - v_2)$ depends solely on the conserved soliton momenta (P_1, P_2) and is independent of the coordinates (X_1, X_2). Since $[\hat{P}_1, \hat{P}_2] = 0$, all operators in $T_{12}(\hat{P}_1, \hat{P}_2)$ mutually commute, and the Jacobian of the Darboux coordinate transformation $X_j \mapsto X'_j = X_j + \gamma_j(P_j)\sum_{i \text{ left of } j} M_i(P_i)$ at fixed (P_1, P_2) is $\det(\partial X'_j/\partial X_k) = 1$. In the field path integral, $T_{12}/\hbar$ is already part of the classical action of the two-soliton saddle point $u_{12}(x,t)$, whose Gaussian fluctuation determinant gives I_{12} / J_{12} directly with no additional $O(\hbar^0)$ cross-term operator-ordering anomaly.*

3. Single-soliton one-loop dispersion and fixed-P^{phys} renormalisation shift: *On the one-soliton background, the one-loop zero-point momentum shift $\Delta P_1(\alpha, \epsilon) = \frac{1}{4\pi}\int_{-\infty}^{\infty} q\, \theta'(q; \alpha, \epsilon)\, dq$ is UV-convergent and evaluates in closed form via $t = \sqrt{1+q^2}$ to*

$$\boxed{\Delta P_1(\alpha, \epsilon) = -\frac{\epsilon}{2\pi}\sum_{r=1}^{3} c_r \sin\phi_r \, \ln\left|\tan\frac{\phi_r}{2}\right|.} \tag{36}$$

Similarly, in box quantisation $q_n L + \theta(q_n) = 2\pi n$ (FK eq. A.6.14), the linear UV divergence $-\frac{3\hbar k_1}{\pi}\sqrt{1+\Lambda^2}$ of the mode energy sum $-\frac{\hbar}{4\pi}\int_{-\Lambda}^{\Lambda}\omega'(q)\theta(q)\,dq$ is exactly cancelled *by the normal-ordering counterterm $+3\langle\phi_x^2\rangle_\Lambda M_1 = +\frac{3\hbar k_1}{\pi}(\sqrt{1+\Lambda^2} - 1)$ of the cubic potential $-\phi_x^3$, leaving the finite closed-form one-loop energy shift*

$$\Delta E_1^{\mathrm{ren}}(\alpha) = \frac{1}{4\pi}\left(-6\sin\alpha - 2\sum_{r=1}^{3} c_r \sin\phi_r \cos\phi_r \, \ln\left|\tan\frac{\phi_r}{2}\right|\right). \tag{37}$$

If the S-matrix is expressed as a function of the ***classical velocity parameters*** *(α_1, α_2), the single-soliton propagator shifts cancel the trajectory shifts at $t = t_0$ (FK [1], eqs. 4.13 and 4.21), so $\delta_1 = \delta_{\mathrm{ph}}$ (I_{12} or J_{12}). If instead the S-matrix is expressed at* ***fixed renormalised physical momenta*** *$P_j^{\mathrm{phys}} = P_j(\alpha_j) + \hbar\,\Delta P_1(\alpha_j, \epsilon_j)$, re-expanding $\delta_0(P_1, P_2)/\hbar$ adds the finite shift*

$$\delta_{\mathrm{renorm}}(\alpha_1, \epsilon_1; \alpha_2, \epsilon_2) = -\sum_{j=1}^{2}\frac{\partial \delta_0}{\partial P_j}\Delta P_1(\alpha_j, \epsilon_j) = \Delta X'_1\, \Delta P_1(\alpha_1, \epsilon_1) + \Delta X'_2\, \Delta P_1(\alpha_2, \epsilon_2). \tag{38}$$

4 Task 3: Comparison with Candidates (A) and (B) in the regular region

Table 1 compares the direct one-loop phase against Candidate (A) (8) and Candidate (B) (9) across five regular overtaking configurations ($\epsilon_1 = \epsilon_2 = +1$, $\alpha_1 < \alpha_2 < \pi/3$) and five regular head-on configurations ($\epsilon_1 = +1, \epsilon_2 = -1$, $\alpha_1 + \alpha_2 < \pi/3$). Following the PI's instruction, we display both I_{12} and J_{12}, and both the fixed-α (FK eq. 4.28) and fixed-$P^{\rm phys}$ conventions.

Table 1: Comparison of the $O(\hbar^0)$ two-soliton scattering phase in the regular region ($\alpha_j < \pi/3$, $A_{12} > 0$) between Candidate (A) ($\frac{\pi}{2}\tau\,\mathrm{sgn}\,\eta$), Candidate (B) (0), and the direct one-loop calculation (**(c)**, check 4 of `phase2/scripts/oneloop_good_twosoliton_phase.py`).

Overtaking collisions ($\epsilon_1 = +1,\ \epsilon_2 = +1,\ \alpha_1 < \alpha_2 < \pi/3 \implies v_1 > v_2,\ \eta < 0,\ \tau = +1$)								
(α_1, α_2)	A_{12}	Cand. (A)	Cand. (B)	$I_{12}^{\rm same}$	$J_{12}^{\rm same}$	$\delta_{\rm renorm}$	$I_{12}^{\rm same} + \delta_{\rm renorm}$	$J_{12}^{\rm same} + \delta_{\rm reno}$
(0.15, 0.30)	0.10610	−1.570796	0	−1.150743	+0.420053	+2.811283	+1.660541	+3.231337
(0.20, 0.60)	0.21918	−1.570796	0	−0.955341	+0.615455	+4.466006	+3.510665	+5.081461
(0.25, 0.50)	0.09715	−1.570796	0	−1.167458	+0.403338	+2.203415	+1.035956	+2.606752
(0.30, 0.80)	0.15509	−1.570796	0	−1.044218	+0.526579	+3.073261	+2.029043	+3.599839
(0.40, 0.65)	0.04202	−1.570796	0	−1.301285	+0.269511	+1.023330	−0.277955	+1.292841
Head-on collisions ($\epsilon_1 = +1,\ \epsilon_2 = -1,\ \alpha_1 + \alpha_2 < \pi/3 \implies A_{12} > 0,\ \eta < 0,\ \tau = -1$)								
(α_1, α_2)	A_{12}	Cand. (A)	Cand. (B)	$I_{12}^{\rm head} = J_{12}^{\rm head}$		$\delta_{\rm renorm}$	$I_{12}^{\rm head} + \delta_{\rm renorm}$	
(0.10, 0.15)	1.04775	+1.570796	0	+0.059626		+5.009121	+5.068747	
(0.20, 0.30)	1.23382	+1.570796	0	+0.171241		+4.188721	+4.359961	
(0.25, 0.45)	1.61562	+1.570796	0	+0.277721		+4.380681	+4.658402	
(0.35, 0.55)	3.23295	+1.570796	0	+0.408191		+4.332791	+4.740982	
(0.40, 0.60)	9.26612	+1.570796	0	+0.478642		+5.241126	+5.719769	

Decisive comparison and implications for the good-equation S-matrix.

1. **Neither Candidate (A) nor Candidate (B) is the exact one-loop S-matrix ((c)):** Both candidates predict a *constant* $O(\hbar^0)$ phase across each kinematic sector ($\mp\pi/2$ for Candidate (A), 0 for Candidate (B)). By contrast, the direct one-loop calculation yields a non-trivial function of (α_1, α_2) with non-vanishing gradient (26)–(27), given in closed form by dilogarithms (32)–(33). Unlike the sine-Gordon model [1] (where the phonon phase-shift integral is proportional to the tree-level phase and is wholly absorbed into the coupling renormalisation $\gamma \to \gamma' = \gamma/(1-\gamma/8\pi)$), the Boussinesq equation is not Lorentz-invariant and the dilogarithm sum $J_{12}(\alpha_1, \alpha_2)$ is *not* proportional to $\delta_0(\alpha_1, \alpha_2)$. Thus a third answer holds at one loop:

$$S(\alpha_1, \epsilon_1; \alpha_2, \epsilon_2) = Z\, \exp\left[\frac{i}{\hbar}\delta_0(\alpha_1, \epsilon_1; \alpha_2, \epsilon_2) + i\,\delta_1(\alpha_1, \epsilon_1; \alpha_2, \epsilon_2) + O(\hbar)\right], \qquad (39)$$

   where $\delta_1 = I_{12}$ (or J_{12} for $Z = e^{i\frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}}$) at fixed (α_1, α_2), plus $\delta_{\rm renorm}$ at fixed $P_j^{\rm phys}$.

2. **What the constant part tells us about Candidate (A) vs. Candidate (B) ((c)/(d)):** Decomposing the raw Faddeev–Korepin integral as $I_{12}^{\rm same} = -\frac{\pi}{2} + J_{12}^{\rm same}(\alpha_1, \alpha_2)$ and $I_{12}^{\rm head} = J_{12}^{\rm head}(\alpha_1, \alpha_2)$ reveals a striking dichotomy between the two channels:

   - In the *overtaking channel* ($\alpha_1 < \alpha_2 < \pi/3$), the $q = 0$ Levinson jump contributes $-\pi/2$, which *matches the $-\pi/2$ Stirling constant of Candidate (A)* ($\frac{\pi}{2}\tau\,\mathrm{sgn}\,\eta = -\frac{\pi}{2}$)! As $\alpha_1 \to \alpha_2^-$ (threshold / equal velocities), $J_{12}^{\rm same}(\alpha_1, \alpha_2) \to 0$ and $I_{12}^{\rm same} \to -\pi/2$.

- In the *head-on channel* ($\epsilon_1 = +1, \epsilon_2 = -1$), the $q = 0$ boundary term (29) vanishes identically because $\theta_1(0^\pm, \epsilon_1 = +1)$ and $\theta_2(0^\pm, \epsilon_2 = -1)$ are supported at opposite signs of $q = 0^\pm$ ($\theta_1(0^+) = 2\pi, \theta_1(0^-) = 0$ whereas $\theta_2(0^+) = 0, \theta_2(0^-) = -2\pi$). Hence there is *no* $+\pi/2$ *constant in the head-on channel*, and as $\alpha_1, \alpha_2 \to 0$, $I_{12}^{\text{head}} = J_{12}^{\text{head}} \to 0$, matching **Candidate (B)** rather than Candidate (A)'s $+\pi/2$!

This explains why the naive analytic continuation of the bad-equation Gamma factor $R(\eta)^{-1}$ to the head-on channel of the good equation failed the defect fusion triangle test by $e^{\pm i\pi/2}$ in `phase2/defect_semiclassics.pdf`: the $q = 0$ zero-mode step is present only when both solitons move in the *same* direction ($\delta_{\epsilon_1,\epsilon_2}$), not with opposite sign in the head-on channel.

5 Task 4: Small-amplitude limit and quantum Toda cross-check

Proposition 5.1 (**(c)**, checks 1–3 of `phase2/scripts/oneloop_good_limits_cusp_resonance.py`)**.** *In the small-amplitude limit $\alpha_1, \alpha_2 \to 0^+$:*

1. Overtaking ($\epsilon_1 = \epsilon_2 = +1$, $\alpha_1 = \lambda\rho, \alpha_2 = \lambda \to 0^+$ at fixed ratio $\rho = \alpha_1/\alpha_2 \in (0, 1)$): *Of the nine terms $(r, s) \in \{1, 2, 3\}^2$ in (32), the eight terms with $(r, s) \neq (3, 3)$ cancel to $O(\lambda^2)$, while the $(3, 3)$ term ($c_3^2 = 4$, $x_{1,3}/x_{2,3} \to \rho$) converges at rate $O(\lambda^2)$ to the scale-invariant KdV dilogarithm function:*

$$\boxed{\lim_{\lambda\to 0^+} J_{12}^{\text{same}}(\lambda\rho, \lambda) = J_{\text{KdV}}(\rho) := \frac{\pi}{2} + \frac{2}{\pi} L(\rho) = \frac{1}{\pi}\big[L(\rho) - L(1/\rho)\big],} \tag{40}$$

and $\lim_{\lambda\to 0^+} I_{12}^{\text{same}}(\lambda\rho, \lambda) = -\frac{\pi}{2} + J_{\text{KdV}}(\rho) = \frac{2}{\pi} L(\rho)$. Notice that $J_{\text{KdV}}(\rho)$ interpolates monotonically between

$$\lim_{\rho\to 1^-} J_{\text{KdV}}(\rho) = 0 \quad \big(I_{\text{KdV}}(1^-) = -\tfrac{\pi}{2}\big), \qquad \lim_{\rho\to 0^+} J_{\text{KdV}}(\rho) = +\frac{\pi}{2} \quad \big(I_{\text{KdV}}(0^+) = 0\big). \tag{41}$$

2. Head-on ($\epsilon_1 = +1, \epsilon_2 = -1$, $\alpha_1, \alpha_2 \to 0^+$): *Every argument $x_{1,r}x_{2,s}$ in (33) either vanishes as $O(\alpha)$ or approaches ± 3 where the leading terms cancel. Expanding to $O(\alpha_1\alpha_2)$ gives the exact asymptotic law*

$$\boxed{I_{12}^{\text{head}}(\alpha_1, \alpha_2) = J_{12}^{\text{head}}(\alpha_1, \alpha_2) = \frac{\alpha_1\alpha_2}{\pi}\left[-3\ln(\alpha_1\alpha_2) + 6\ln 2 - \frac{3}{2}\ln 3 - 3\right] + O(\alpha^4 \ln\alpha) \xrightarrow{\alpha_j\to 0} 0.} \tag{42}$$

3. Single-soliton dispersion as $\alpha \to 0^+$: *Because the phonon group velocity $v_g(q) = \text{sgn}(q)(1 + 2q^2)/\sqrt{1+q^2}$ satisfies $|v_g(q)| > 1 > |\cos\alpha| = |v_1|$ for all $q \neq 0$,* no phonon ever co-moves with a good-Boussinesq soliton. *Consequently, there is no co-moving phonon capture resonance (unlike the $q^* = -\sqrt{3}/2$ resonance of the bad equation in* ***`phase2/oneloop.pdf`*** *Prop. 2.2), and $\Delta P_1, \Delta E_1^{\text{ren}}$ vanish smoothly as $\alpha \to 0^+$:*

$$\Delta P_1(\alpha, +1) = \frac{\alpha}{\pi}\left(\ln\frac{\alpha}{2} - 1 + \frac{\ln 3}{4}\right) + O(\alpha^3 \ln\alpha) \to 0. \tag{43}$$

Moreover, in the Hirota delay shift $-\ln(A_{12})\big(\frac{\Delta P_1(\alpha_1)}{k_1} - \frac{\Delta P_1(\alpha_2)}{k_2}\big)$ for overtaking collisions, the $\ln\lambda$ and constant terms cancel identically between the two solitons, leaving $-\frac{2}{\pi}\ln\rho\,\ln\big|\frac{1-\rho}{1+\rho}\big|$, which precisely cancels the logarithmic part of $\frac{2}{\pi}L(\rho)$ in (31) to leave the pure Legendre chi function $-\frac{4}{\pi}\chi_2(\rho)$.

Cross-check with the quantum Toda Fermi-edge route (`phase2/toda_route.pdf`, `phase2/toda_edge.pdf`). In `phase2/toda_edge.pdf`, the small-amplitude double-scaling limit ($\beta_j = \sqrt{y_j \hbar_B} \to 0$ at fixed y_j) of the quantum Toda Bethe ansatz established two facts:

1. For *head-on* scattering, the exact Toda phase equals $\delta_0/\hbar$ with no non-trivial $O(\hbar^0)$ two-body residual. Proposition 5.1(2) ((42): $I_{12}^{\text{head}} = O(\alpha_1\alpha_2 \ln(\alpha_1\alpha_2)) \to 0$) is in *complete agreement* with this head-on Toda limit **(c)**.

2. For *overtaking* scattering, `toda_edge.pdf` normalised the phase by subtracting the Fermi-edge value $y_2 \to 0^+$ ($\arg S(1,2) - \arg S(1,e_2)$) and found that Candidate (A) is very accurate when $y_1 \gg y_2$ ($\rho = \sqrt{y_2/y_1} \ll 1$), but leaves a genuine two-body residual of $O(10^{-1})$ depending on *both* y_1 and y_2 when $\rho = \sqrt{y_2/y_1}$ is closer to 1 (e.g. $(y_1, y_2) = (1.5, 1), (3, 1), (1.2, 1)$). Equation (40)–(41) exhibits the semiclassical counterpart of this two-body structure **(c)**/**(d)**: $I_{\text{KdV}}(\rho) = \frac{2}{\pi} L(\rho)$ vanishes as $\rho \to 0^+$ (when one soliton approaches the Fermi edge e_2) and reaches $-\frac{\pi}{2}$ as $\rho \to 1^-$ (when the two amplitudes become comparable).

6 Task 5: Behaviour near the cusp and near the resonance lines

6.1 Near the cusp $\alpha \to \pi/3$

Proposition 6.1 (**(c)**, check 4 of `phase2/scripts/oneloop_good_limits_cusp_resonance.py`)**.** *Let $\alpha_2 = \frac{\pi}{3} - \varepsilon$ ($\varepsilon \to 0$) at fixed $\alpha_1 \in (0, \pi/3)$:*

1. Classical degeneracy and Darboux pole: *As $\varepsilon \to 0$, $\frac{dP_2}{d\alpha_2} = \sqrt{3}\,\epsilon_2 \varepsilon + O(\varepsilon^2) \to 0$ and changes sign across the cusp, so the effective inverse mass $\frac{dv_2}{dP_2} = -\frac{1}{2\varepsilon} + O(1)$ diverges and flips sign. While the Hirota peak shift $\Delta X_2 = \frac{\ln A_{12}}{k_2}$ remains finite at $\alpha_2 = \pi/3$, the Darboux factor $\gamma_2 = \frac{\sqrt{3}}{\varepsilon} + O(1)$ and the canonical shift $\Delta X_2' = -\frac{\partial \delta_0}{\partial P_2}$ (hence also δ_{renorm}) have a* simple pole $\propto (\frac{\pi}{3} - \alpha_2)^{-1}$.

2. Detachment of the instability mode and $\pm\pi$ jump of I_{12}: *At $\alpha_2 = \frac{\pi}{3} - \varepsilon$, the first root angle $\phi_{2,1} = \alpha_2 + \frac{2\pi}{3} = \pi - \varepsilon$ approaches π, so the Lorentzian width $|\sin\phi_{2,1}| = \sin\varepsilon \approx |\varepsilon|$ in (19) shrinks to zero at $q = 0$ and $\tan(\phi_{2,1}/2) \approx 2/\varepsilon \to \pm\infty$. Consequently:*

   - $\Delta P_1(\alpha_2)$ *is continuous at $\alpha_2 = \pi/3$ with a vertical cusp derivative $\partial_{\alpha_2} \Delta P_1(\alpha_2) \sim \frac{\epsilon_2}{2\pi} \ln|\frac{\pi}{3} - \alpha_2|$;*
   - $I_{12}^{\text{same}}(\alpha_1, \alpha_2)$ *is continuous across $\alpha_2 = \pi/3$ (since $\tan(\phi_{1,r}/2)/\tan(\phi_{2,1}/2) \to 0$ where $L(0) = 0$), with logarithmic derivative singularity $\partial_{\alpha_2} I_{12}^{\text{same}}(\alpha_1, \alpha_2) \propto \ln|\frac{\pi}{3} - \alpha_2|$;*
   - $I_{12}^{\text{head}}(\alpha_1, \alpha_2)$ *and $I_{12}^{\text{same}}(\alpha_2, \alpha_1)$ have $\tan(\phi_{2,1}/2)$ in the numerator, which flips from $+\infty$ ($\varepsilon > 0$) to $-\infty$ ($\varepsilon < 0$). Since $L(+\infty) - L(-\infty) = -\frac{\pi^2}{2}$ and $\sum_{r=1}^{3} c_r \operatorname{sgn}(\tan(\phi_{1,r}/2)) = -2$, both jump discontinuously by $+\pi$ across the cusp:*

$$\lim_{\varepsilon \to 0^+} \Big[I_{12}^{\text{head}}\big(\alpha_1, \tfrac{\pi}{3} - \varepsilon\big) - I_{12}^{\text{head}}\big(\alpha_1, \tfrac{\pi}{3} + \varepsilon\big) \Big] = \lim_{\varepsilon \to 0^+} \Big[I_{12}^{\text{same}}\big(\tfrac{\pi}{3} - \varepsilon, \alpha_1\big) - I_{12}^{\text{same}}\big(\tfrac{\pi}{3} + \varepsilon, \alpha_1\big) \Big] = +\pi. \tag{44}$$

   Beyond the cusp ($\alpha_2 > \pi/3$), the one-soliton is linearly unstable ($\tilde{\omega} = \pm i\gamma(\alpha_2)$, Proposition 2.4), so the oscillatory one-loop fluctuation integral around it acquires an imaginary instability width and an asymptotic one-soliton scattering state no longer exists.

6.2 Near the resonance lines ($A_{12} \to 0$ or ∞)

The classical two-soliton solution encounters a Y-junction resonance whenever two KP roots coincide:

(i) *Head-on* $A_{12}^{\text{head}} \to \infty$*:* $p_{1b} = p_{2a}$, i.e. $\alpha_1 + \alpha_2 = \frac{\pi}{3}$ (which forms the upper boundary of the regular head-on triangle $\alpha_1 + \alpha_2 < \frac{\pi}{3}$ inside $(0, \pi/3)^2$);

(ii) *Overtaking* $A_{12}^{\text{same}} \to \infty$ *and head-on* $A_{12}^{\text{head}} \to 0$*:* $p_{1a} = p_{2b}$ or $p_{1b} = p_{2a}$, i.e. $|\alpha_1 - \alpha_2| = \frac{\pi}{3}$;

(iii) *Overtaking* $A_{12}^{\text{same}} \to 0$*:* $p_{1a} = p_{2a}$, i.e. $\alpha_1 + \alpha_2 = \frac{2\pi}{3}$ (mirror image across the cusp).

Proposition 6.2 (**(c)**, check 5 of `phase2/scripts/oneloop_good_limits_cusp_resonance.py`)**.** *On all three resonance lines (i)–(iii):*

1. Finiteness and C^1-smoothness of I_{12} and J_{12} at fixed (α_1, α_2): *On each resonance line, one of the nine root pairs* (r, s) *satisfies* $\sin^2 \phi_{1,r} = \sin^2 \phi_{2,s}$*, so the argument* $z_{r,s}$ *of* $L(z_{r,s})$ *in* (32)–(33) *hits* $z_{r,s} = \pm 1$*. Because* $L'(z) = \frac{2 \ln |z|}{1 - z^2}$ *has a* removable singularity *at* $z = \pm 1$ *(*$\lim_{z \to \pm 1} L'(z) = -1$*), both* I_{12}, J_{12} *and their first derivatives* $\partial_{\alpha_j} I_{12}, \partial_{\alpha_j} J_{12}$ *(where the vanishing denominator* $\sin^2 \phi_{2,s} - \sin^2 \phi_{1,r}$ *in* (34) *is cancelled by the numerator) remain* completely finite and continuous *right onto the resonance lines! For example, on the regular-region head-on resonance line* $\alpha_1 + \alpha_2 = \pi/3$ *at* $\alpha_1 = 0.4, \alpha_2 = \pi/3 - 0.4 \approx 0.647198$*,* I_{12}^{head} *approaches the finite limit* $+0.5164972485$ *linearly in* $h = \frac{\pi}{3} - \alpha_1 - \alpha_2$*.*

2. Logarithmic divergence of delays and δ_{renorm}, and breakdown of the saddle point for $A_{12} < 0$: *On every resonance line,* $\ell_1 = \ell_2$ *(*$\eta = 0$*), so the tree-level phase* $\delta_0 = \frac{2}{9}(\ell_1 - \ell_2) \ln A_{12} + T_{12} \to T_{12}$ *stays finite while the classical spatial shift* $\Delta X_j = \mp \ln(A_{12})/k_j$*, the Darboux shift* $\Delta X_j'$*, and hence the momentum-renormalisation shift* δ_{renorm} (38) diverge logarithmically as $\ln |A_{12}|$*. Physically, as* $A_{12} \to 0$ *or* ∞*, the two solitons fuse into the resonant third soliton of their KP triple for a time* $\Delta t \propto |\ln A_{12}| \to \infty$*; and in the singular region* $A_{12} < 0$ *beyond the resonance line,* $\tau_{12}(x, t)$ *vanishes along a spacetime curve where* $u_{12} = 2(\log \tau_{12})_{xx}$ *has a* $-2/(x - x_0(t))^2$ *pole, so the regular two-soliton saddle point ceases to exist.*

7 Summary of claims, labels, and checks

Claim	Label	Verification script & check
Quartic co-moving dispersion has 2 real propagating roots and 1 evanescent pair for all $\tilde{\omega} \neq 0$	**(c)**	`oneloop_good_phonon_scattering.py` (check 1)
Exact Hirota linearised eigenmode solves fourth-order PDE $H_u \delta u = 0$ with $R(q) = 0$ and $C_\pm(q) = 0$	**(c)**	`oneloop_good_phonon_scattering.py` (check 2)
KP root form (15) and unitarity $\lvert A(1, \mathrm{ph}(q))\rvert = 1$ on $\mathbb{R} \setminus \{0\}$	**(c)**	`oneloop_good_phonon_scattering.py` (check 3)
Three-root arctan formula (18) for $\theta(q)$ and rational/algebraic formula (19) for $\theta'(q)$	**(c)**	`oneloop_good_phonon_scattering.py` (check 3)
Levinson mode count $N_{\mathrm{Lev}} = 1$ for $\alpha < \pi/3$ and $N_{\mathrm{Lev}} = 2$ for $\alpha > \pi/3$	**(c)**	`oneloop_good_phonon_scattering.py` (check 4)
Discrete modes from KP triple: zero mode $K = k_1$ and heavy-branch instability $\gamma = \sqrt{3}(\sin^2\alpha - 3/4)$ for $\alpha > \pi/3$	**(c)**	`oneloop_good_phonon_scattering.py` (check 5)
Exact two-soliton phonon eigenmodes (25), $\hat{c} = 0$, and factorised transmission $A(1, \mathrm{ph})A(2, \mathrm{ph})$	**(c)**	`oneloop_good_twosoliton_phase.py` (check 1)
UV convergence of I_{12}, J_{12}, equality $\nabla I_{12} = \nabla J_{12} = \nabla \delta_{\mathrm{ph}}$, and $I_{12} = J_{12} - \frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}$	**(b)**/**(c)**	`oneloop_good_twosoliton_phase.py` (check 2)
Closed-form Legendre-χ_2 formulas (32)–(33) and crossing $I_{12}^{\mathrm{head}}(\alpha_1, \alpha_2) = I_{12}^{\mathrm{same}}(\alpha_1, \pi - \alpha_2)$	**(c)**	`oneloop_good_twosoliton_phase.py` (check 2)
Closed-form $\Delta P_1(\alpha)$ (36), normal-ordered $\Delta E_1^{\mathrm{ren}}(\alpha)$ (37), and δ_{renorm} (38)	**(c)**	`oneloop_good_twosoliton_phase.py` (checks 3–4)
Maslov index $\mu_{\mathrm{Maslov}} = 0$ and absence of $O(\hbar^0)$ cross-term ordering anomaly in regular region	**(c)**	`oneloop_good_twosoliton_phase.py` (§3)
Direct one-loop phase is a non-constant function of (α_1, α_2), ruling out both (A) and (B) as exact	**(c)**	`oneloop_good_twosoliton_phase.py` (check 4, Table 1)
Small-amplitude overtaking limit $J_{12}^{\mathrm{same}} \to J_{\mathrm{KdV}}(\alpha_1/\alpha_2)$ (40) and head-on limit $I_{12}^{\mathrm{head}} = O(\alpha_1\alpha_2 \ln(\alpha_1\alpha_2)) \to 0$ (42)	**(c)**	`oneloop_good_limits_cusp_resonance.py` (checks 1–3)
Cusp $\alpha \to \pi/3$: $dP/d\alpha \to 0$, simple pole of $\Delta X_j'$ and δ_{renorm}, and $\pm\pi$ jump (44) of I_{12}	**(c)**	`oneloop_good_limits_cusp_resonance.py` (check 4)
Resonance lines $A_{12} \to 0, \infty$: removable singularity of $L(z)$ at $\lvert z\rvert = 1$, finite I_{12}, J_{12}, and $\ln\lvert A_{12}\rvert$ divergence of δ_{renorm}	**(c)**	`oneloop_good_limits_cusp_resonance.py` (check 5)

Failures recorded.

1. In an initial numerical test of the Levinson integral $\int_0^\infty \theta'(q)\, dq$, we differentiated $\ln A(1, \mathrm{ph}(q))$ numerically with `mpmath.diff` inside `mpmath.quad`. Because `mpmath.diff` uses complex steps $q + he^{i\phi}$, the branch condition `if q >= 0` inside `phonon(q)` failed off the real axis and returned spurious integrals of order 10^{21}. Replacing numerical differentiation by the exact rational/algebraic expression (19) restored 35-digit accuracy.

2. In an initial test of the hyperbolic formula for $\partial_u \theta(-\sinh u; \alpha, +1)$ $(q < 0)$, we used the wrong sign in the continuation $\alpha \mapsto \pi - \alpha$ (which swaps the KP roots $p_a \leftrightarrow -p_b$ and adds a minus sign), producing an $O(1)$ mismatch (0.501) for $q < 0$. Tracking the $p_a \leftrightarrow -p_b$ swap fixed the sign and gave agreement to 3.7×10^{-37}.

3. In an initial numerical check of the normal-ordered energy integral $-\int_0^\infty [\omega'(q)(\theta(q) - \theta(-q)) - 12k_1 q/\sqrt{1+q^2}]\,dq$ using `mpmath.quad(..., [0, 1, 5, 20, mp.inf])`, evaluating at the deepest tanh-sinh nodes $q \sim 10^{35}$ caused catastrophic cancellation between p_{1b} and $\sqrt{1+q^2}/(2\sqrt{3})$ in $z_b(q)$, returning $\sim 4 \times 10^{36}$. Cancelling the $-6k_1$ tadpole algebraically in the rational integrand (37) eliminated the subtraction completely and gave agreement with the closed form to 7.5×10^{-37}.

4. When testing the small-amplitude head-on asymptotic coefficient of $\frac{\alpha_1\alpha_2}{\pi}$ in (42), we initially counted two terms instead of four in the $\{1,2\} \times \{1,2\}$ block (omitting $(1,2)$ and $(2,1)$ at $z=-3$, where $[zL'(z)]'$ is even and equals $+\frac{5\ln 3-4}{16}$ just as at $z=+3$), leaving a constant offset of $\frac{5\ln 3-4}{4} \approx 0.373265$. Including all four terms gave the exact constant $6\ln 2 - \frac{3}{2}\ln 3 - 3$, reducing the residual to $O(\alpha^4 \ln \alpha)$ (2.2×10^{-18} at $\lambda = 10^{-5}$).

5. In an early check of the Darboux shift $\Delta X_j' = -\partial\delta_0/\partial P_j$, we wrote $\frac{dP}{d\alpha} = 2\epsilon \sin^2\alpha \cos 3\alpha$ (which vanishes at $\alpha = \pi/6$) instead of $\frac{dP}{d\alpha} = \frac{2}{3}\epsilon \sin\alpha \sin 3\alpha$ (which vanishes at the cusp $\alpha = \pi/3$). Cross-checking against `corrigan-zambon/defect_two_soliton_symplectic.py` caught the error and yielded $\Delta X_j' + \partial\delta_0/\partial P_j = 0$ to 35 digits. Also, in the first run of `phase2/scripts/oneloop_good_phonon_scattering.py`, a typo ($-6\partial_\xi^2(u_0 f)$ instead of $+6\partial_\xi^2(u_0 f)$ in the static co-moving operator check) caused the assertion to fail with residual 1.61; fixing the sign reduced the residual to 3.76×10^{-37}.

6. In the first run of `phase2/scripts/oneloop_good_twosoliton_phase.py`, `delta_renorm_good` defined δ_{renorm} as $-\sum_j \Delta X_j' \Delta P_j$ while the numerical check evaluated $-\sum_j (\partial\delta_0/\partial P_j)\Delta P_j = +\sum_j \Delta X_j' \Delta P_j$ (since $\Delta X_j' = -\partial\delta_0/\partial P_j$ already carries the minus sign), triggering the sign assertion in check 4. Aligning the sign to $-\sum_j (\partial\delta_0/\partial P_j)\Delta P_j = +\sum_j \Delta X_j' \Delta P_j$ passed all checks to 10^{-32}.

References

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